

Gabriel Pedde Ungureanu | Physics

PhD | HPC & AI Master

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Profile

Mathematician (110/110 *cum laude*) and PhD candidate in high energy physics **based in Brescia**. I work where the mathematics behind cryptography — number theory, finite fields, probability — meets low-level computer engineering that decides whether an implementation is secure, from writing cache- and timing-aware C/C++ down to Bayesian Inference applied to HPC and Machine Learning problems.

Research & Engineering Experience

SISSA

Trieste, Italy

Doctoral Researcher — Theoretical Physics & Scientific Computing

2022–2026

- Exact, non-perturbative results in quantum field theory via large-scale **symbolic and numerical computation** — finite-field / modular arithmetic, combinatorial algorithms, generating functions — on SISSA's Ulysses EuroHPC cluster; **6 published papers**, ~60 citations.
- Engineered a verification-first pipeline where every candidate result is refereed by a **deterministic symbolic/numerical evaluator** (never self-graded), recovering exact solutions to **machine precision**.

Education

SISSA / ICTP Joint Programme

Trieste, Italy

Master in High Performance Computing and AI (MHPC)

2025–2026

Concurrent with the last PhD year. Delved into **Computer architecture** (memory hierarchy, pipelines, caches), **quantum computing** (gate model, Grover, error correction), low-level C/C++, and machine learning.

Università Cattolica del Sacro Cuore

Brescia, Italy

BSc Mathematics & MSc Physics (Laurea Triennale & Laurea Magistrale)

2017–2022

Number theory, abstract algebra, probability, real & functional analysis. MSc thesis in **quantum complexity, holography, information theory**, published in Physical Review D.

Awards & Publications

6 peer-reviewed papers in theoretical physics (~60 citations) — [INSPIRE-HEP](#) | [Google Scholar](#)

110/110 *cum laude* — MSc Physics (**best graduate of the year**) | 110/110 *cum laude* — BSc Mathematics | Istituto Toniolo merit fellowship

Selected Projects — Security-Relevant Foundations

Signal-in-noise: Built a quasi-periodic **Gaussian Process from scratch** (no libraries): Cholesky log-marginal-likelihood maximisation, spectral init, calibrated posterior | [GitHub](#)

Microarch. & timing: Hands-on experiments with cache hierarchies, branch prediction, cycle-level timing, memory-access patterns — benchmarked on Leonardo & LUMI | [GitHub](#)

Low-level systems: Performance-critical **C / C++ / CUDA** from first kernels to a full Lattice-Boltzmann solver; MPI + cuBLAS linear algebra. Correct, fast, resource-constrained code | [GitHub](#)

Technical Skills

Crypto foundations: Number theory & modular arithmetic, finite fields, probability, information theory, quantum information (Shor/Grover, post-quantum)

Information Theory: Statistical signal extraction, hypothesis testing, Bayesian inference, cache/timing microarchitecture, constant-time reasoning

Low-level programming: C, C++, Fortran; CPU microarchitecture (memory hierarchy, pipelines), cache timing analysis, Linux

Languages/tools: Python (NumPy/SciPy), Mathematica, Bash, Git; PyTorch, scikit-learn; CUDA, MPI, Docker

Spoken languages: Italian (native) | English (C1) | Romanian (A2)

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